

4.6.7. If k is an integer, what is $\lceil k + (1/2) \rceil$? Why?
 Suppose k is an integer.

We must show the ceiling of $k + (1/2)$ and explain.

Given any real number x , the ceiling of defined as follows:

$$\lceil x \rceil = n \leftrightarrow n - 1 < x \leq n$$

Let $x = k + (1/2)$, for some $k \in \mathbb{Z}$

Let's start by finding a pattern:

$$k = -1$$

$$x = -1 + (1/2) = -0.5$$

$$\lceil x \rceil = \lceil -0.5 \rceil = 0$$

$$k = 0$$

$$x = 0 + (1/2) = 0.5$$

$$\lceil x \rceil = \lceil 0.5 \rceil = 1$$

$$k = 1$$

$$x = 1 + (1/2) = 1.5$$

$$\lceil x \rceil = \lceil 1.5 \rceil = 2$$

$$x = k + (1/2)$$

$$\lceil x \rceil = \lceil k + (1/2) \rceil = k + 1$$

Let $n = k + 1$, for some $k \in \mathbb{Z}$

$$\lceil x \rceil = n \leftrightarrow n - 1 < x \leq n$$

$$\lceil k + (1/2) \rceil = (k + 1) \leftrightarrow (k + 1) - 1 < (k + (1/2)) \leq (k + 1)$$

$$= k + 1 \leftrightarrow k < (k + (1/2)) \leq k + 1$$

Solution: The ceiling of $k + (1/2)$ is $k + 1$, because $k < k + (1/2) \leq k + 1$,

by definition of ceiling, $\lceil k + (1/2) \rceil = k + 1$.